GuitarTap

Tap Tone Analysis Report

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Measurement Name: Plate UMIK-1 3 Tap Web Mac
Type: Material (Plate)
Frequency Range: 20.0 Hz – 200.0 Hz
Microphone: Umik-1 Gain: 18dB · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
36.4 Hz	-53.9 dB	D1	8.3	Diagonal (fLC)
67.5 Hz	-52.8 dB	C#2	11.5	Longitudinal (fL)
117.4 Hz	-52.1 dB	A#2	26.7	Cross-grain (fC)

Sample Dimensions

Length: **557.50 mm** Width: **220.50 mm** Thickness: **4.85 mm**
Mass: **208.0 g** Density: **0.349 g/cm³**

Body Dimensions

Body Length (a): **490 mm** Lower Bout Width (b): **368 mm**
Panel Stiffness: **f_vs = 60 (Classical Top)**

Gore Target Thickness

2.78 mm

Plate Properties

Speed of Sound (L): **4209 m/s**
Speed of Sound (C): **1146 m/s**
Young's Modulus (L): **6.18 GPa**
Young's Modulus (C): **0.46 GPa**
GLC (Shear Modulus): **0.360 GPa**
Cross/Long Ratio: **0.074**
typical: 0.04–0.08

Specific Modulus (L): **17.7** (Fair)
Specific Modulus (C): **1.3** (Very Good)
Radiation Ratio (L): **12.1**
Radiation Ratio (C): **3.3**
Long/Cross Ratio: **13.5**
typical: 12–25

Overall Quality: **Good**

Three-Tap Measurement Process:

- **1. Longitudinal (fL) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (fC) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. Diagonal (fLC) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.